

Haotian Lin

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EDUCATION

TSINGHUA UNIVERSITY

B.S. in Mechanics & B.E. in Vehicle Engineering

Beijing

9/2020 – 6/2024

- GPA: 3.6/4.0
- Visiting research intern at University of California, Berkeley

CARGENIE MELLON UNIVERSITY

M.S. in Robotics

Pittsburgh

8/2024– now

- GPA: 4.0/4.0
- Related Courses: Advanced Machine Learning(A), Optimal Control & RL(A), Introduction to Robot Learning(A)

RESEARCH EXPERIENCE

LeCAR Lab & Auton Lab (Carnegie Mellon University)

Pittsburgh, PA

Research Assistant (MSR)

Advisor: [Guanya Shi](#) (guanyas@andrew.cmu.edu) , [Jeff Schneider](#) (jeff4@andrew.cmu.edu)

Improving Model-Based RL for Continuous Control

10/2024 – 3/2024

- Uncover and study structural off-policy issue in plan-based MBRL.
- Propose a minimalist fix to mitigate value oversativate in TD-MPC family and boost performance for high-dimensional continuous control tasks.

Self-improving Foundation Models via Residual RL

5/2025 – 10/2025

- Building data-efficient RL pipeline for manipulation in JAX for real-world learning (YAM, Franka) and simulation benchmarks (LIBERO, SimplerEnv, Mimicgen), supports data collection through different teleoperation approaches and self-play, integrates pipeline for VLA SFT and evaluation.
- Leverage expressive policy prior (pi-0, octo, OpenVLA-oft) to warm-start exploration and probing using a residual RL. Employ a hybrid rollout scheme to collect diverse recovery behavior and distilled these curated behavior back into the generalist via SFT.

Multi-task RL for Robotics in Parallel Simulation

5/2025–Now

- Study gradient conflict and task representation learning for value-based on-policy algorithm in multi-task learning.

Mechanical System Control Lab (University of California, Berkeley)

Berkeley, CA

Research Assistant

Advisor: [Masayoshi Tomizuka](#) (tomizuka@berkeley.edu)

Diffusion Model for Multi-Agent Pedestrian Trajectory Prediction

10/2023 – 4/2024

- Leverage latent diffusion to model complex social interaction between agents. Propose a novel trajectory prediction scheme that consider trajectory prediction as inpainting problem and solved through posterior sampling provide
- Contribute to conference papers admitted to IROS 2024.

Prediction Benchmark

7/2023 – 11/2023

- Evaluate prediction model's ability of interaction modeling and map representation, bridging the gap of concept interpretation and model performance by Shapley Value-based contribution metrics.

THICV (Tsinghua University)

Beijing

Research Assistant

Advisor: Jianqiang Wang (wjqlws@tsinghua.edu.cn) , Shaobing Xu (shaobxu@tsinghua.edu.cn)

GNN-based prediction framework for generalizable scene understanding

10/2022 – 6/2023

- Designing social interaction aware backbone leveraging prior knowledge for autonomous driving, adaptable to downstream tasks including trajectory prediction and intention detection. Designing conditional diffusion model-based traffic simulator that integrates natural language commands.
- Contribute to conference papers admitted to IEEE IV 2023 and IROS 2025.

iDLab (Tsinghua University)

Beijing

Research Assistant

(Students Research Training Program)

Advisor: [Shengbo Li \(lishbo@tsinghua.edu.cn\)](mailto:lishbo@tsinghua.edu.cn)

DRL Algorithm Development for Integrated Decision-Making & Control

10/2022 – 6/2023

- Testing and verification of **GOPS** toolkit in driving scenario.
- Develop offline RL algorithm for diffusion and flow policy using Q-score alignment.

PUBLICATIONS

- Wenli Xiao, **Haotian Lin**, Haoru Xue, Tairan He, Zhengyi Luo, Yuqi Xie, Fengyuan Hu, Andy Peng, Guanya Shi, Jim Fan, Yuke Zhu. "Self-Improving Vision-Language-Action Models with Data Generation via Residual RL" (In Submission)
- **Haotian Lin**, Pengcheng Wang, Jeff Schneider, Guanya Shi. "TD-M (PC) $\hat{S}^2$: Improving Temporal Difference MPC Through Policy Constraint". arXiv preprint arXiv:2502.03550. (In Submission)
- **Haotian Lin**, Yixiao Wang, Mingxiao Huo, Chensheng Peng, Zhiyuan Liu, Masayoshi Tomizuka. "Joint Pedestrian Trajectory Prediction through Posterior Sampling". 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems.
- Zhiyuan Liu, Leheng Li, Yuning Wang, **Haotian Lin**, Hao Cheng, Zhizhe Liu, Lei He, Jianqiang Wang. "Controllable traffic simulation through llm-guided hierarchical chain-of-thought reasoning". 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems.
- Yuning Wang, Zhiyuan Liu, **Haotian Lin**, Jinhao Li, Ruochen Li, Jianqiang Wang. "Vif-gnn: A novel agent trajectory prediction model based on virtual interaction force and gnn". 2023 IEEE Intelligent Vehicles Symposium (IV). IEEE, 2023.

SKILLS & INTERESTS

- Strong interest in **data-driven reinforcement learning, generative models for control**.
- Coding skills: Python (PyTorch, JAX), C++, Matlab.
- Robotic Physical Simulator: IsaacSim, IsaacGym, Mujoco
- Software: SolidWorks, Abaqus

HONOR

Xingjian Yingcai Scholarship, Tsinghua University

2022

Xingjian Yingcai Scholarship, Tsinghua University

2021

EXTRACURRICULAR EXPERIENCES

Secondary Award in Xindong Project (Intelligent vehicle design contest, Tsinghua University)

1/2021

Member of Tsinghua Photography Team (official art group in Tsinghua University)

9/2021 – 6/2024

Committee Member of Tsinghua Photography Team

9/2022 – 6/2023

Member of Student Union (Department Of Art) of Xingjian College

9/2021 – 6/2022